FINANCIAL RISK ANALYSIS

Market Risk Monitoring Using Statistical Models

Executive Report

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Scope:
Analysis of extreme events and dynamic risk estimation in equity markets using volatility modeling and event intensity frameworks.

Executive Summary

This report presents a data-driven framework to monitor financial risk in equity markets through the identification and modeling of extreme events. Using historical data from the S&P 500 index, we estimate market volatility and the intensity of extreme negative returns.

The analysis shows that periods of elevated volatility are strongly associated with an increased frequency of extreme events. This relationship enables the construction of a dynamic risk indicator capable of identifying high-risk regimes.

Key Insight: Market risk evolves dynamically and increases significantly during periods of high volatility.

1 Market Data Captures Long-Term Trends and Extreme Behavior

The analysis is based on daily historical data from the S&P 500 index, including price levels, returns, and derived risk indicators.

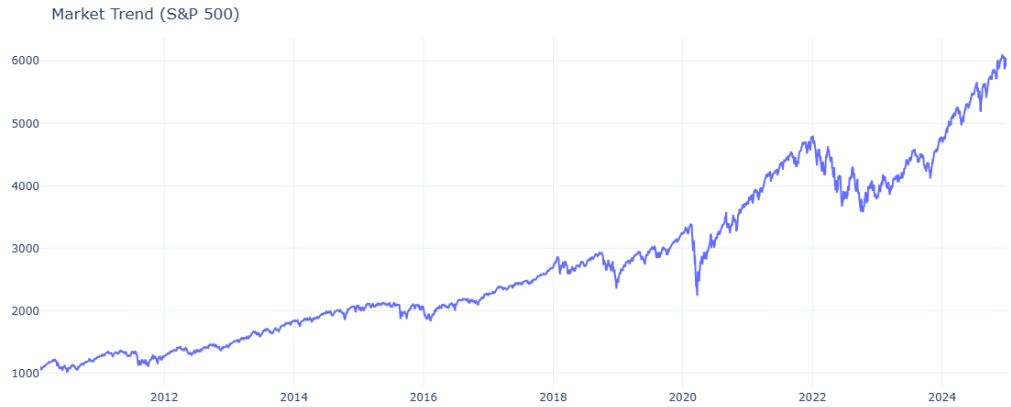

Figura 1: S&P 500 price evolution showing long-term market trends

Interpretation: The market exhibits long-term growth punctuated by periods of instability, where abrupt declines reflect systemic stress.

2 Extreme Events Can Be Systematically Identified

Extreme events are defined as significant negative returns below a percentile-based threshold.

Interpretation: Extreme events tend to cluster during periods of market instability.

3 Volatility Dynamics Reveal Risk Accumulation

Market volatility is modeled using a GARCH framework.

Interpretation: Volatility clustering indicates persistence in market instability.

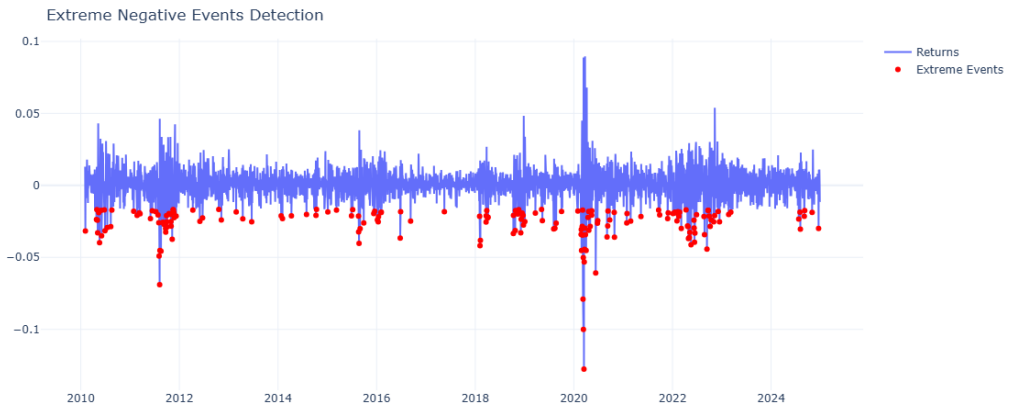

Figura 2: Extreme negative returns identified using a percentile-based threshold

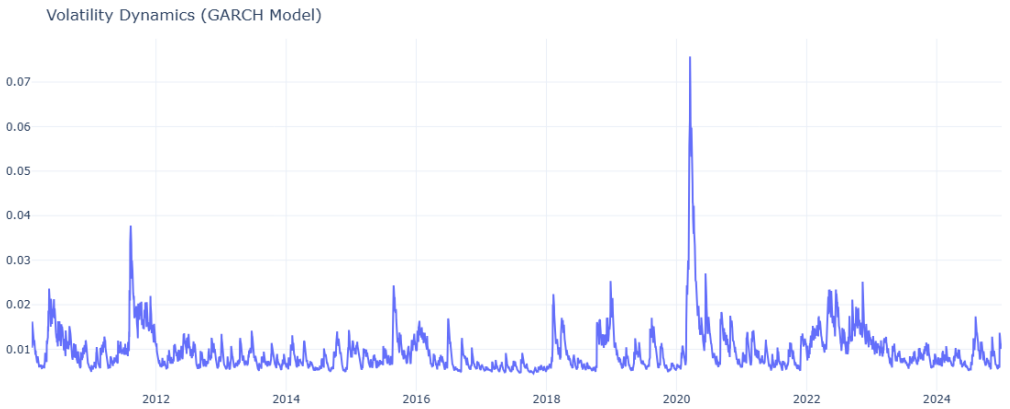

Figura 3: Time-varying volatility estimated using a GARCH model

Insight: Sustained increases in volatility are strong indicators of upcoming clusters of extreme events.

4 Risk Intensity Increases During Market Stress Periods

The intensity function $\lambda(t)$ represents the expected frequency of extreme events.

Interpretation: Elevated values of $\lambda(t)$ indicate periods of heightened systemic risk.

Insight: The intensity function $\lambda(t)$ acts as a forward-looking indicator of financial stress.

5 Interpretation of Risk Intensity

The intensity function $\lambda(t)$ provides a direct and interpretable measure of risk.

Higher values indicate increased likelihood of extreme events, while lower values reflect stability.

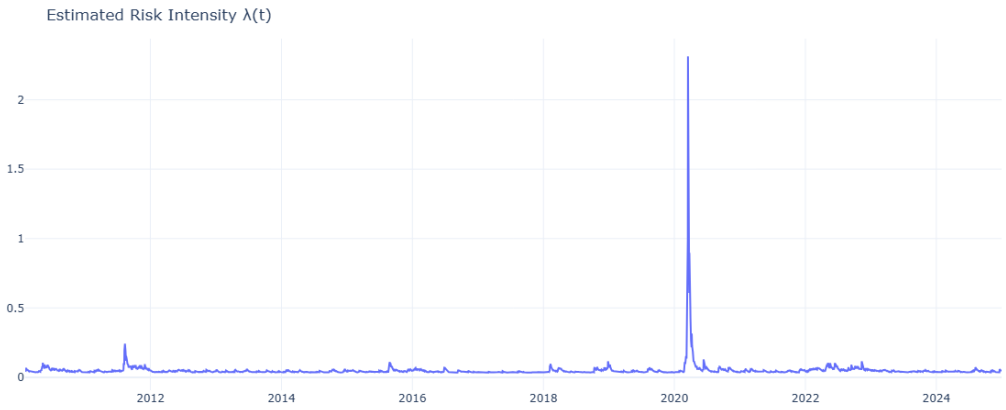

Figura 4: Estimated intensity of extreme events representing dynamic market risk

6 Business Impact and Practical Applications

The proposed framework enables:

- Real-time risk monitoring
- Early detection of market stress
- Improved portfolio management
- Identification of high-risk regimes

7 Limitations of the Approach

- Based on historical data
- Does not include macroeconomic shocks
- Simplified modeling assumptions

8 Future Improvements and Extensions

- Inclusion of macroeconomic variables
- Multivariate models
- Machine learning integration
- Real-time deployment

9 Conclusion

This report demonstrates that financial risk can be effectively monitored using volatility and event-based modeling.

Final Insight: Combining volatility modeling with event intensity provides a robust framework for anticipating financial risk.